

SMART INDIA HACKATHON 2024



- **Problem Statement ID - SIH1788**
- **Problem Statement Title - Development and Implementation of Face Recognition Technology in the Police Department**
- **Theme - Smart Automation**
- **PS Category - Software**
- **Team ID - 33276**
- **Team Name - BLACK-SQUAD**





SMART CRIME DETECTION



❖ Proposed Solution

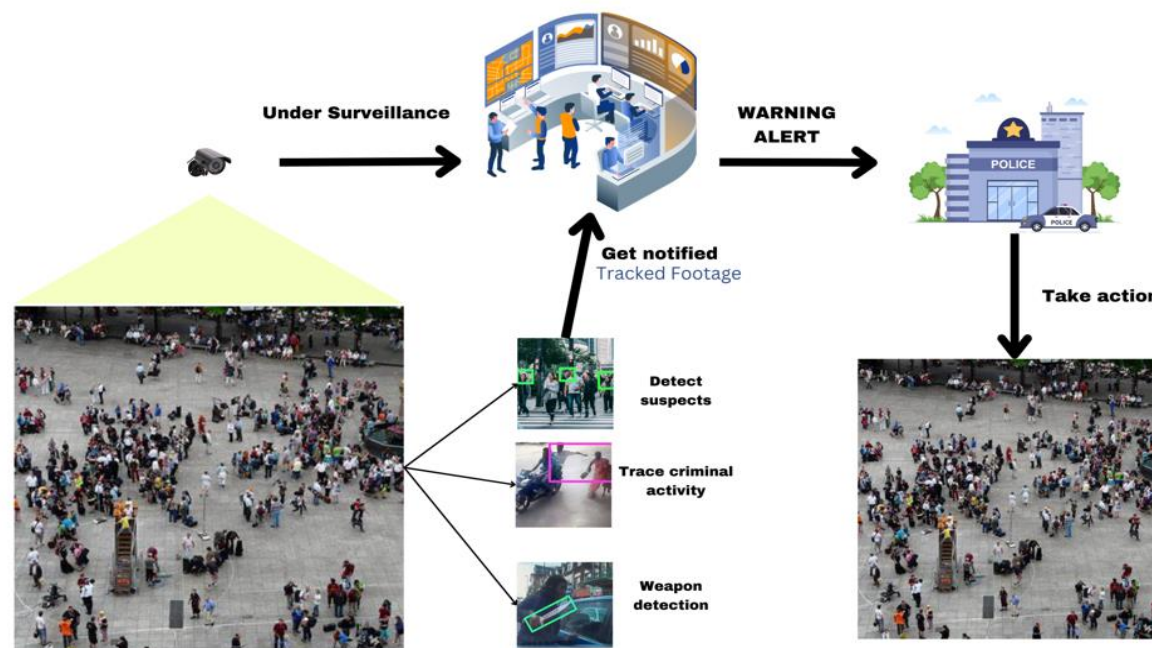
The police department is enhancing public safety with a Face Recognition System (FRS) designed to Identify suspects, Identify missing persons, Track criminal activities and Record evidence footage, Detect unusual behavior , and Spot dangerous weapons through CCTV

❖ How it addresses the problem

- Criminal Detection and Tracking
- Identity missing persons in crowd
- Weapon and Suspicious Behavior Detection
- Control Room Alerts and Police Response

❖ Innovation and uniqueness of the solution

- Real-Time Criminal Tracking
- Integrated Multi-Layer Detection
- Instant Notifications to Control Centers
- Behavioral Anomaly Detection
- Weapon Detection and Instant Alerts
- Adaptive AI
- Automated Evidence Capture
- Scalability





TECHNICAL APPROACH

❖ Technologies to be used:

Hardware Requirements:

- **Networking:** Routers, switches.
- **Edge Computing:** Edge processing devices.
- **Additional Hardware:** GPU, CCTV, HDD/SSD

Software Requirements:

- **Programming Languages:** Python
- **Frame Works:** PyTorch, OpenCV, NumPy, YOLOv5
- **Storage Technologies:** Database, Cloud Storage

❖ Methodology and process for implementation:

Scope: Define objectives and requirements

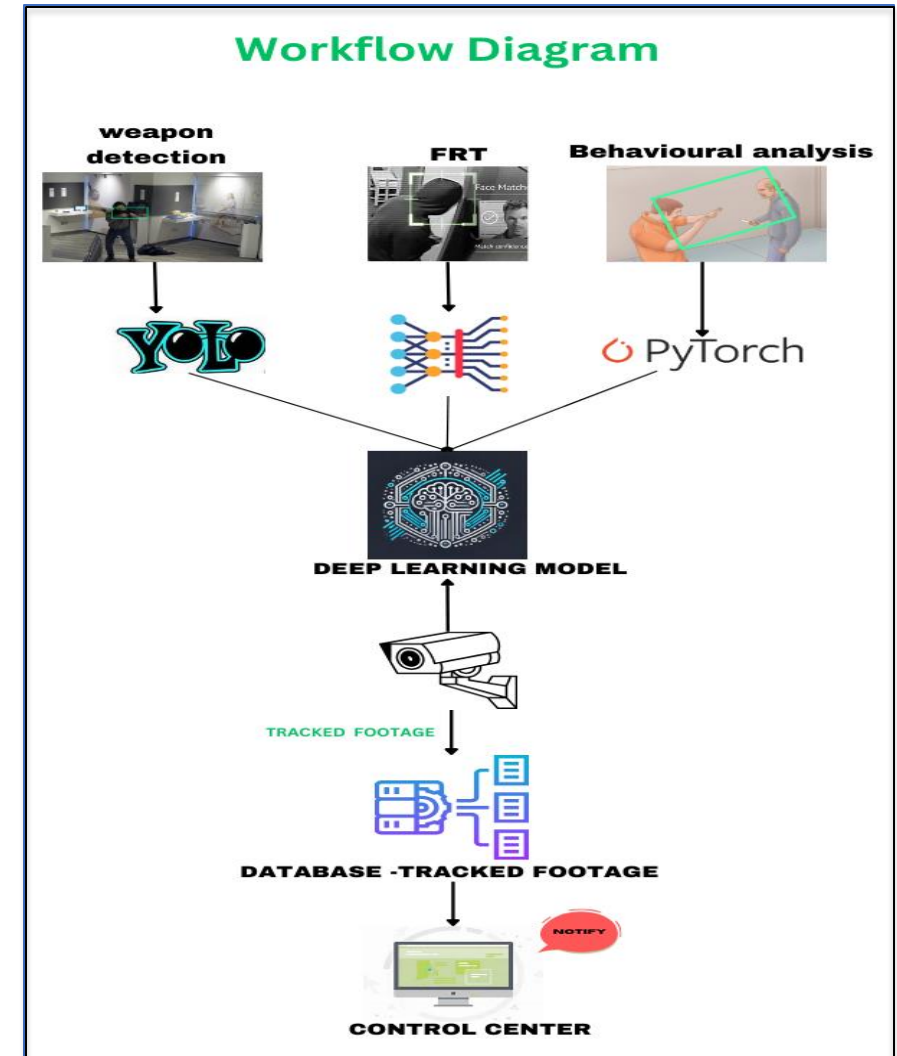
Design: Create system architecture diagrams.

Development: Train YOLOv5 model with data.

Deployment: Install and configure system on-site.

Maintenance: Monitor and update system regularly.

- ❖ **Product status:** 60% product build is completed and further build is in progress.





FEASIBILITY AND VIABILITY



❖ Feasibility:

Growing Demand: Increased need for advanced surveillance in public safety.

Real -Time Crime Prevention: Instant detection of suspects, weapons, and criminal activity.

Scalability: Applicable to various sectors beyond law enforcement.

❖ Challenges:

Technical: Accuracy issues, system integration with current infrastructure.

Ethical/Privacy: Data privacy concerns, risk of bias, public backlash.

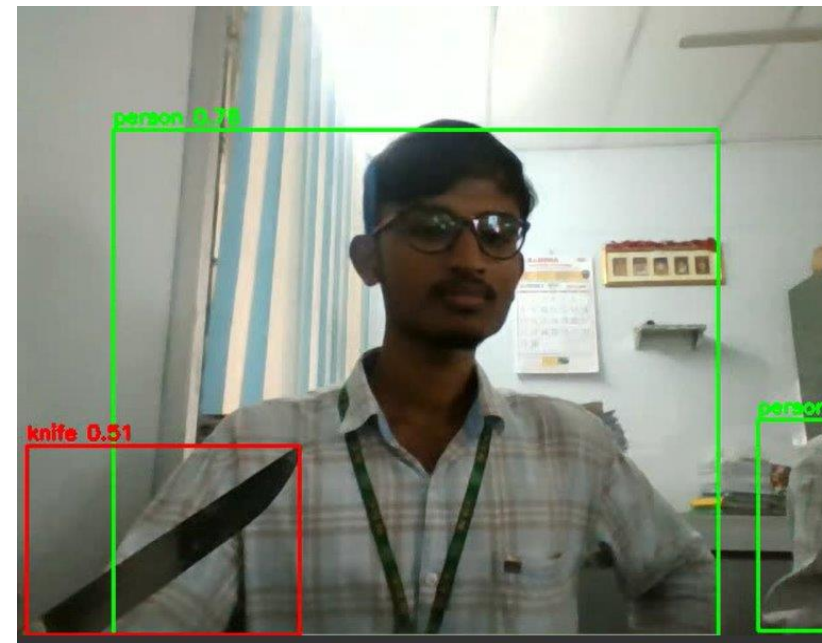
Legal: Regulatory hurdles and compliance with privacy laws.

❖ Strategies:

Improve AI: Adaptive AI for enhanced accuracy and reduced false positives.

Ethical Safeguards: Transparent communication, data encryption, and limited access.

Legal Compliance: Collaboration with legal experts, advocacy for responsible frameworks.



[Weapon Demonstration](#)



IMPACT AND BENEFITS

❖ Potential impact on the target audience:

Enhanced Public Safety through real-time criminal detection.

Faster Law Enforcement Response with instant alerts.

Increased Public Confidence in security measures

❖ Benefits of the solution:

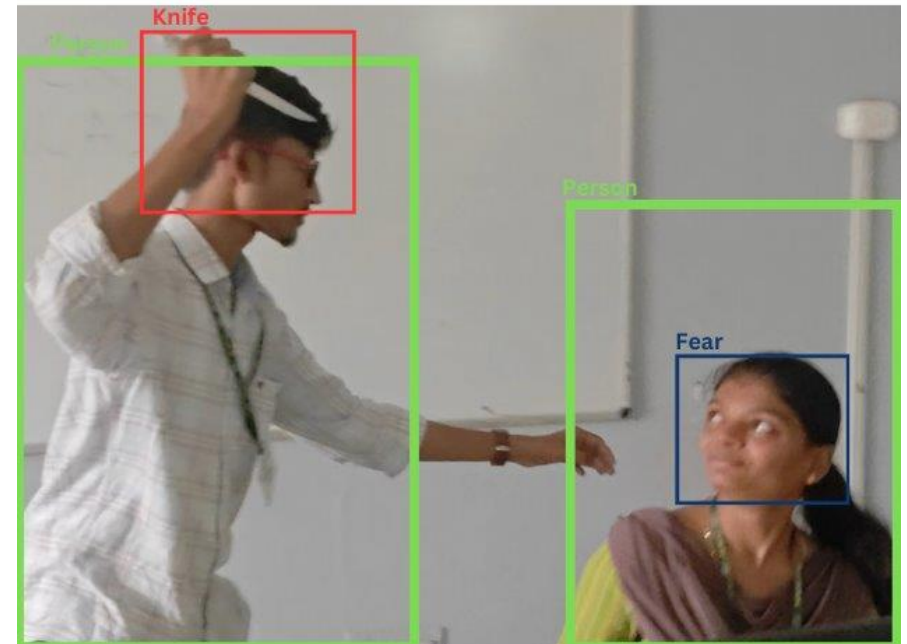
Social: Safer public spaces, reunification of missing persons.

Economic: Reduced crime-related costs, boosted economic activity.

Environmental: Efficient use of resources, minimized crime-related damage.

❖ Channels:

- Department of Police
- Law Enforcement and public safety
- Airports and Transportation
- Event Management
- Government /Private institutions



Detection Demonstration

YOUTUBE:

<https://youtube.com/@black-squad-x6l?si=UQ3b8vIBJRXrOvQz>



RESEARCH AND REFERENCES



Surveillance Technology:

Growing use of facial recognition for real-time criminal identification. NIST highlights accuracy improvements but acknowledges challenges like poor lighting and demographic bias.[NIST Face Recognition Vendor Test (FRVT)](<https://www.nist.gov/programs-projects/face-recognition-vendor-test-frvt>)

Ethical Concerns:

Privacy risks and demographic biases raise concerns, with organizations like EFF calling for stricter regulations. [EFF: Face Recognition](<https://www.eff.org/pages/face-recognition>)

AI for Behavior & Weapon Detection:

AI's potential in detecting behavioral anomalies and weapons, but false positives remain a challenge(<https://www.cmu.edu/news/stories/archives>)

Market Trends:

The facial recognition market is expected to grow significantly, with multi-layer detection systems gaining traction. [Allied Market Research - Facial Recognition Market](<https://www.alliedmarketresearch.com/facial-recognition-market>)

High Accuracy:

CNN-based facial recognition systems can achieve up to 99.38% accuracy in real-time suspect identification(<https://www.researchsquare.com/article/rs-3596384/v1>).

Deep Learning Models:

Systems like FaceNet and VGGface are effective in criminal detection, especially in low-light or occluded conditions, enhancing overall robustness.(<https://link.springer.com>).

Vulnerability to Attacks:

Face recognition systems face challenges such as morphing attacks, necessitating the development of advanced algorithms to counter these vulnerabilities. (<https://link.springer.com/article/10.1007/s00145-022-09312-1>).

Behavioral Detection:

The integration of behavioral anomaly detection enhances real-time monitoring capabilities, allowing for timely alerts and police response(<https://www.sciencedirect.com/science/article/pii/S0167739X2031177X>).